

Aathish . S

3122218001001

UCB2501 - COMPUTER NETWORKS

ASSIGNMENT - I -

$$1) \text{ Bandwidth} = 20 \text{ Mbps} \\ = 20 \times 10^3 \text{ bps} \\ = 20,000 \text{ bps}$$

$$\text{Throughput} = \frac{20,000 \times 14,000}{60} = 4.67 \text{ Mbps}$$

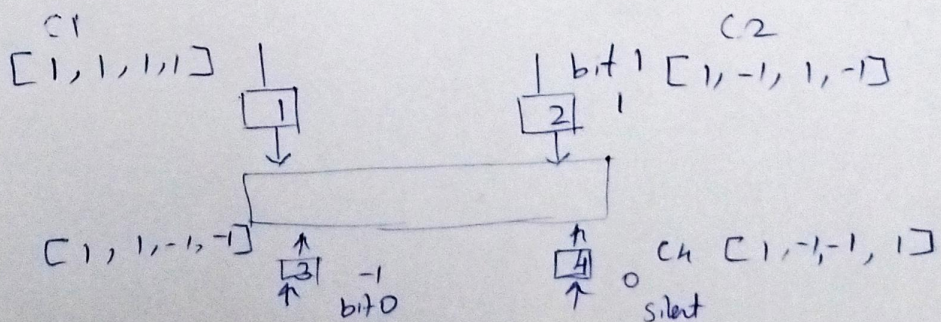
$$2) \text{ Propagation time} = \frac{\text{Distance}}{\text{propagation speed}} = \frac{18,000 \times 1000}{2.4 \times 10^8 \text{ m/s}} \\ = 0.075 \text{ s}$$

$$3) \text{ Propagation time} = \frac{\text{Distance}}{\text{propagation speed}} = \frac{19000 \times 1000 \text{ m}}{2.4 \times 10^8 \text{ m/s}} = 0.0792 \text{ s}$$

$$4) \text{ Transmission time} = \frac{\text{Message time}}{\text{bandwidth}} = \frac{5500 \text{ bytes}}{2 \times 10^9 \text{ bytes/s}} = 22 \times 10^{-6} \\ = 22 \mu\text{s}$$

$$4) \text{ Time taken} = \frac{\text{No of frames}}{\text{bandwidth}} = \frac{200,000}{15000 \text{ bps}} = 13.33 \text{ s}$$

5) Station 1 $\rightarrow$ databit 1 $\rightarrow$ 1
Station 2 $\rightarrow$ databit 1 $\rightarrow$ 1
Station 3 $\rightarrow$ databit 0 $\rightarrow$ -1
Station 4 $\rightarrow$ silent $\rightarrow$ 0



$$d_1 \cdot c_1 = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$$

$$d_2 \cdot c_2 = \begin{bmatrix} 1 & -1 & 1 & -1 \end{bmatrix}$$

$$d_3 \cdot c_3 = \begin{bmatrix} -1 & -1 & 1 & 1 \end{bmatrix}$$

$$d_4 \cdot c_4 = \begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 3 & 1 \end{bmatrix}$$

Data on the common channel $D = [1, -1, 3, 1]$

If st. 1 wants to listen to st. 2 then

$$\text{st. 2} \rightarrow [1, -1, 1, -1]$$

$$= [1, -1, 3, 1] [1, -1, 1, -1]$$

$$= [1, 1, 3, -1]$$

$$= 4/4 = 1$$

Station 3 listens to station 4

$$\text{Station 4} \rightarrow [1, -1, 1, 1]$$

$$= [1, -1, 3, 1] [1, -1, 1, 1]$$

$$= [1, 1, 3, 1]$$

$$= 0/4$$

$$= 0 \text{ (silent)}$$

6.

$$m = \text{length of data} = 8$$

$$2^n \geq m+r$$

$$\text{Data} \rightarrow 11011001$$

To find r : -

$$2^r \geq m+r$$

When $r=0$:

$$2^0 \neq 8+0$$

When $r=1$

$$2^1 \neq 8+1$$

When $r=2$

$$2^2 \geq 8+2$$

$$4 \geq 10$$

When $r=3$

$$2^3 \geq 8+3$$

$$8 \geq 11$$

When $r=4$

$$2^4 \geq 8+4$$

$$16 \geq 12$$

$$\text{So } r=4$$

4 redundant bits

$$\therefore \text{Total} = 8+4 = \underline{\underline{12}}$$

1	1	0	1	r_8	1	0	0	r_4	1	r_2	r_1
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To find r_1

$r_1 \rightarrow$ Check 1 bit, Skip 1 bit

$r_1 \rightarrow 11101r_1$

$r_1 \rightarrow 0$

$r_2 \rightarrow$ Check 2 bit, Skip 2 bit...

$r_2 \rightarrow 10101r_2$

$r_2 \rightarrow 1$

$r_2 = 0$

$r_4 \rightarrow$ Check 4 bit, skip 4 bit

$r_4 \rightarrow 1100r_4$

$r_4 = 0$

$r_8 \rightarrow$ Check 8 bit skip 8 bit

$r_8 \rightarrow 1101r_8$

$r_8 = 1$

Redundant $\rightarrow r_8 r_4 r_2 r_1 \rightarrow 1010$

Encapsulated Data

1	1	0	1	1	1	0	0	0	1	1	0
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Introduce error in the 6th bit

1	1	0	1	1	1	1	0	0	1	1	0
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$$r_1 = 111010$$

$$r_1 = 0$$

$$r_2 = 101111$$

$$r_2 = 1$$

$$r_4 \rightarrow 11100$$

$$r_4 = 1$$

$$r_8 = 11011$$

$$r_8 = 0$$

$$\text{Redundant Bits} = r_8 r_4 r_2 r_1 = 0110$$

Error exists in Bit $0110_2 = b_{10}$

Flip bit b_{10}

Corrected:

1	1	0	1	1	1	0	0	0	1	1	0
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